

RESEARCH OBJECTIVE

Improve the **robustness, stability, and efficiency** of dynamic graph networks operating under **three conditions**, while satisfying **four requirements**

R1: Robustness

R2: Accuracy

R3: Efficiency

R4: Privacy



C1: Adversarial Environment



C2: Distributed Dynamics



C3: Non-Stationarity

● **M1: CT-TGNN**

● **M5: Stoch. Transformer**

● **M4: MambaShield**

○ **M7: Certification**

● **M1: CT-TGNN**

● **M3: TripleE-TGNN**

● **M4: MambaShield**

● **M2: FedLLM-API**

● **M6: UC-HGP**

● **M5: Stoch. Transformer**

● **M4: MambaShield**

○ **M7: Certification**

All 12 condition–requirement intersections are covered by M1–M7



APPLICATION DOMAIN (Chapter 3): Network Security • Healthcare • Finance • Autonomous Systems • Agriculture

M1: CT-TGNN
(Neural ODE)

M3: TripleE-TGNN
(Multi-level)

M5: Stoch. Transformer
(Uncertainty-aware)

M2: FedLLM-API
(Federated)

M4: MambaShield
(SSM)

M6: UC-HGP
(Bayesian)

M7: Certification
(Game-Theoretic)



Primary method (main contributor)



Supporting method (secondary role)